Topic 6 — Electricity, Magnetism and Electromagnetism

MANDATORY SYLLABUS SCOPE: Covers all dimensions of Electric Current, Potential Difference, Ohm's Law, Resistance and Resistivity, Circuits (Series vs Parallel), Electric Power and Commercial Energy Units, Safety Devices (Fuse wires, Earthing, 3-pin plug), AC vs DC Generation, Transformers and Rectifiers, Lighting Technologies (Incandescent, Fluorescent, CFL, LED, Sodium lamps), Magnetism & Electromagnetism, Renewable Solar/Wind Energy, EV Battery Cathode Chemistries, and Distributed Energy Resources. Fully synchronized with Ghatna Chakra Physics, NCERT Class 10–12, and latest 2021–2025 State PCS PSC papers.

Current Affairs (2024–2026 Focus)

1. **Electric Vehicle (EV) Battery Cathode Chemistries (IAS Pre 2025 Fact):** - Lithium-ion batteries used in EVs comprise four primary parts: Cathode, Anode, Electrolyte, and Separator. - **Cathode (Positive Electrode):** Typically composed of **Cobalt, Lithium, and Nickel** (as NMC: $\mathrm{LiNiMnCoO_2}$) or Lithium Iron Phosphate (LFP: $\mathrm{LiFePO_4}$), coated on aluminium foil. - **Anode (Negative Electrode):** Constructed of **Graphite (Carbon)** coated on copper foil. - **Common Trap Fact:** Graphite is strictly an **ANODE** material, NOT a cathode component!

2. **Alternative Powertrain Vehicles (IAS Pre 2025 Fact):** - Refers to propulsion systems replacing conventional internal combustion engines (petrol/diesel). - Classified into: (i) Full Battery Electric Vehicles (BEVs), (ii) Hydrogen Fuel Cell Vehicles (FCEVs), and (iii) Fuel Cell-Electric Hybrids. All three represent alternative powertrains.

3. **Distributed Energy Resources (DER: IAS Pre 2024 Fact):** - Decentralized, small-scale power generation situated near consumption sites:
    1. Rooftop solar photovoltaic units
    2. Battery storage systems
    3. Biomass generators (utilizing agricultural/waste gas)
    4. Fuel cells and mini-hydro schemes

4. **Pumped-Storage Hydropower (PSH) in Uttar Pradesh (IAS Pre 2024 Fact):** - MoEFCC approved massive PSH projects ($11.98\ \mathrm{GW}$), including Greenko's $3.66\ \mathrm{GW}$ **project in Sonbhadra District, Uttar Pradesh**. PSH functions as a giant grid-scale battery by pumping water to upper reservoirs during low-tariff hours and releasing it through turbines during peak demand.

5. **Silkyara Tunnel Lighting Calculation (UPPCS RO/ARO Pre 2023 Fact):** - 15 bulbs of $200\ \mathrm{W}$ burning continuously for $24\ \mathrm{hours}$ consume:

$$\text{Total Energy} = \frac{15 \times 200\ \mathrm{W} \times 24\ \mathrm{h}}{1000} = 72\ \mathrm{kWh\ (units)}$$

6. **Wind Power Installed Capacity in India (CGPSC / MPPSC Fact):** - India ranks **4th globally** in wind power capacity (after China, USA, Germany). - Total capacity crossed $56\ \mathrm{GW}$. - State rankings:

Gujarat is #1 ($\sim 27.9\%$, over $15.6\ \text{GW}$), having overtaken **Tamil Nadu (#2**, $\sim 21.7\%$, $12.1\ \text{GW}$). Madhya Pradesh ranks 7th.

7. **Global Solar Silicon Wafer Production (IAS Pre Fact):** - Global fab capacity: **China ranks #1**, followed by Taiwan (#2) and South Korea (#3). - In India, solar tariffs are regulated by **CERC and SERCs** via competitive bidding (SECI is a trading licensee, not tariff setter).

Detailed In-Depth Core Theory

6.1 Charge, Current, Potential and Circuit Laws

1. **Current and Current Density:** - Electric current is the macroscopic charge flow rate: $I = dq/dt$. - Current density $\vec{j}$ represents the microscopic distribution: $I = \int \vec{j} \cdot d\vec{A}$. In terms of carrier charge e, number density n, and drift velocity $\vec{v}_d$: $\vec{j} = ne\vec{v}_d$.
2. **Ohm's Law & Resistance:** - Under constant physical conditions, $V \propto I \implies V = IR$. - Resistivity $\rho = m/(ne^2\tau)$, where m is electron mass and τ is relaxation time. - Temperature dependence in metals: $\rho_T = \rho_0[1 + \alpha(T - T_0)]$, where $\alpha > 0$ (resistance rises with temperature due to increased electron-phonon scattering).
3. **Power and Joule Heating:** - Heating effect: $H = I^2Rt = \frac{V^2}{R}t = VIt$. - Power: $P = VI = I^2R = V^2/R$. - For household appliances in parallel ($V = \text{constant}$), $P \propto 1/R$. Thus, a $40\ \text{W}$ bulb has higher resistance than a $100\ \text{W}$ bulb.

6.2 Household Safety, Earthing and Safety Engineering

1. **Electric Fuse Wire:** - Placed in the live wire before appliances. - Operating requirement: Must melt and isolate the circuit prior to current damaging appliance wiring or starting a fire. - Made of eutectic Tin-Lead alloy ($63\%\ \text{Sn} + 37\%\ \text{Pb}$), melting point 183°C. High resistivity ensures rapid local heating (I^2R) during fault.
2. **Three-Pin Plug & Earthing:** - The earth pin is longer to ensure protective grounding is established prior to live contact. It has a larger diameter to prevent accidental insertion into the live or neutral socket.
3. **Short Circuit Physics:** - Direct connection between live and neutral wires reduces circuit resistance to almost zero, causing current $I = V/R \to \infty$, generating extreme instantaneous heat.

6.3 Electromagnetic Induction, AC vs DC and Transformers

1. **Faraday's Laws of Induction:** - Induced EMF $\varepsilon = -N\frac{d\Phi_B}{dt}$. - Negative sign represents Lenz's Law (induced current opposes the flux change causing it).
2. **AC vs DC Machinery:** - AC Generator uses slip rings to output alternating sinusoidal current $V = V_0 \sin(\omega t)$. - DC Dynamo uses a split-ring commutator to invert every negative half-cycle into unidirectional pulsing DC.
3. **Transformers:** - Operates on mutual induction across iron core: $\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$. - **Direct current (DC) produces zero time-varying magnetic flux** ($d\Phi/dt = 0$), so transformers CANNOT function on DC. Connecting DC will overheat and burn the primary winding.

6.4 Clean Energy Transitions, Battery Storage & Electric Mobility

1. **Photovoltaic Solar Energy:** - Silicon solar cells exploit the photovoltaic effect: incident photons with energy $h\nu > E_g$ generate electron-hole pairs across a p-n junction, creating a DC voltage ($0.5 - 0.6\ \text{V}$ per cell). - Net metering enables bidirectional grid exchange.
2. **Distributed Energy Resources (DER):** - Incorporates rooftop solar, battery energy storage systems (BESS), biomass generation, and fuel cells situated adjacent to consumer load centres.
3. **Electric Vehicle Batteries:** - Lithium-ion cells use intercalation chemistry:
   - Cathode: Transition metal oxides (Cobalt, Lithium, Nickel, Manganese) on aluminium.
   - Anode: Graphitic carbon on copper.

6.5 Electrical Measuring Instruments & Electromagnetic Induction Devices

1. **Galvanometer**: - Detects the presence and direction of minute electric currents. - Operates on the principle of magnetic torque exerted on a current-carrying coil placed in a magnetic field ($\tau = NIAB$).
2. **Ammeter**: - Measures electric current in amperes (A). - **Connection**: Connected in **SERIES** with the circuit load so that the entire current passes through it. - **Resistance**: An ideal ammeter has **ZERO internal resistance** ($R_A = 0$) so it does not alter circuit current. - **Conversion**: A moving-coil galvanometer is converted into an ammeter by connecting a very low resistance (called a **Shunt**, S) in **PARALLEL** across the galvanometer coil.
3. **Voltmeter**: - Measures electrical potential difference between two points in volts (V). - **Connection**: Connected in **PARALLEL** across the circuit component being measured. - **Resistance**: An ideal voltmeter has **INFINITE internal resistance** ($R_V = \infty$) so it draws zero current from the branch. - **Conversion**: A moving-coil galvanometer is converted into a voltmeter by connecting a very high resistance (R) in **SERIES** with the galvanometer coil.
4. **Devices Operating on Electromagnetic Induction (Faraday's Law)**: - **Dynamo / Electric Generator**: Converts mechanical energy to electrical energy via electromagnetic induction. - **Transformer**: Steps up or steps down alternating voltage via mutual electromagnetic induction. - **Electric Motor**: Converts electrical energy to mechanical rotational motion via electromagnetic forces/induction. - **Induction Cooktop**: Generates heat in ferromagnetic pans via induced eddy currents (I^2R). - *Trap Alert*: Moving-coil ammeters and voltmeters operate on magnetic deflection torque ($\tau = NIAB$), NOT on electromagnetic induction.

Bilingual Terminology -- Electricity and Magnetism

English Term	Hindi Term	Key Anchor
Electric Charge (विद्युत आवेश)	विद्युत आवेश	Fundamental property; positive (proton) or negative (electron); SI unit = Coulomb
Electric Current (विद्युत धारा)	विद्युत धारा	Flow of electric charge; I = Q/t; SI unit = Ampere (A)
Voltage / EMF (विभवान्तर / विद्युत वाहक बल)	विद्युत विभव	Electric potential difference; drives current; SI unit = Volt (V)

English Term	Hindi Term	Key Anchor
Resistance (प्रतिरोध)	प्रतिरोध	Opposition to current flow; R = V/I; SI unit = Ohm (omega)
Ohm's Law (ओम का नियम)	ओम का नियम	V = IR; current proportional to voltage at constant temp
Conductors (चालक)	विद्युत चालक	Allow electricity to flow; metals, graphite
Insulators (विद्युतरोधी)	विद्युतरोधी	Resist current; rubber, glass, wood, plastic
Semiconductors (अर्धचालक)	अर्धचालक	Between conductor and insulator; silicon, germanium; basis of electronics
Power (शक्ति)	शक्ति	P = VI = I^2 R = V^2/R; SI unit = Watt (W)
Magnet (चुम्बक)	चुम्बक	Object that produces magnetic field; attracts iron, nickel, cobalt
Electromagnetism (विद्युत चुम्बकत्व)	विद्युत चुम्बकत्व	Current produces magnetic field; basis of motors, generators
Faraday's Law (फैराडे का नियम)	फैराडे का नियम	Changing magnetic flux induces EMF; basis of generators/transformers
Transformer (ट्रांसफार्मर)	परिणामित्र	Step-up or step-down AC voltage; works on mutual induction
AC / DC (प्रत्यावर्ती / दिष्ट धारा)	AC / DC	AC = alternating current (direction changes); DC = direct current
Earthing (भूसंपर्क)	अर्थिंग	Safety connection to ground; prevents electric shock
Fuse (फ्यूज)	फ्यूज	Safety device; thin wire that melts on excess current (made of tin-lead alloy)

One-Liner Revision -- Electricity and Magnetism

#	Fast Fact
1	Charge on electron = -1.6 x 10^-19 C; proton = +1.6 x 10^-19 C
2	1 Coulomb = charge of 6.24 x 10^18 electrons
3	Ohm's Law: V = IR (Voltage = Current x Resistance)
4	Resistance of wire: R = rho x L / A (longer and thinner = higher resistance)
5	Series circuit: same current; resistances add (R_total = R1 + R2)
6	Parallel circuit: same voltage; conductances add (1/R_total = 1/R1 + 1/R2)
7	Power consumed: P = VI = I^2 R = V^2/R
8	Household supply (India) = 220V, 50Hz AC
9	Fuse wire = tin-lead alloy; low melting point; melts to break circuit on overload

#	Fast Fact
10	MCB (Miniature Circuit Breaker) = modern fuse; can be reset; trips on overload
11	Earth wire = green/yellow; safety; carries fault current to earth
12	Neutral wire = blue/black; Live/Phase wire = red/brown
13	Short circuit = live and neutral wires touch directly; large current; fire risk
14	Fleming's Left Hand Rule = direction of force on current-carrying conductor in magnetic field
15	Fleming's Right Hand Rule = direction of induced EMF in generator
16	Faraday's Law: induced EMF proportional to rate of change of magnetic flux
17	Lenz's Law: induced current opposes the change causing it (conservation of energy)
18	Transformer: step-up (V2>V1, N2>N1); step-down (V2<V1, N2<N1)
19	Electric motor: converts electrical energy to mechanical energy
20	Generator/Dynamo: converts mechanical energy to electrical energy
21	AC generator uses **slip rings**; DC generator uses **split-ring commutator**
22	Magnetic poles: like poles repel, unlike poles attract
23	Earth's magnetic pole (north seeking) = geographic south pole of Earth
24	Magnetic declination = angle between geographic north and magnetic north
25	Solenoid with current = acts as bar magnet; strength increases with turns and current
26	Electromagnet = soft iron core + coil; temporary; used in cranes, MRI, doorbells
27	MRI (Magnetic Resonance Imaging) = uses strong magnetic field + radio waves
28	Lightning conductor = iron rod on tall buildings; safely channels lightning to earth
29	Electric bell works on: **electromagnetic induction + make-and-break circuit**
30	Galvanometer = detects small currents; can be converted to voltmeter (series R) or ammeter (parallel R)